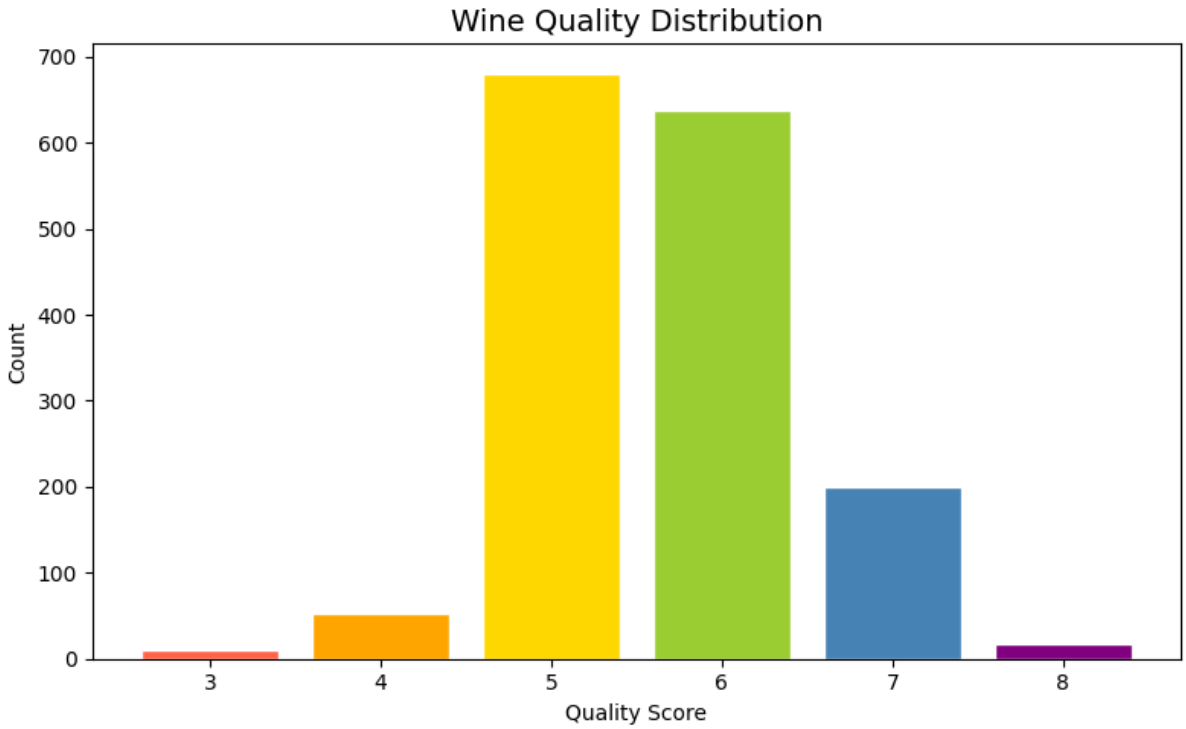
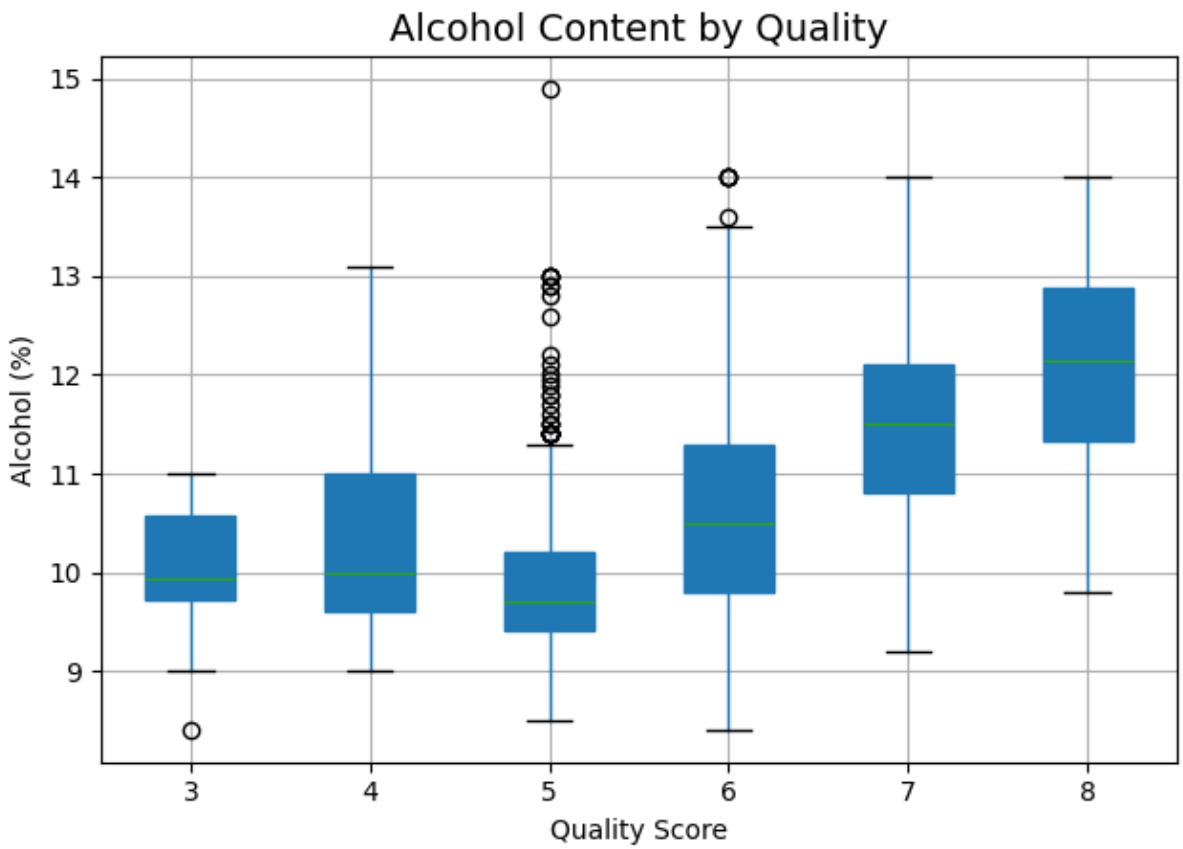
```
# — 1. LOAD DATA ————————————————————————————————————

url <- "https://archive.ics.uci.edu/ml/machine-learning-databases/wine-quality/winequality-red.csv"

wine <- read.csv(url, sep = ";")


# — 2. GRAPH 1: Quality Distribution (Bar Plot) ————————————

barplot(table(wine$quality),

      main = "Wine Quality Distribution",

      xlab = "Quality Score",

      ylab = "Count",

      col  = c("tomato","orange","gold","yellowgreen","steelblue","purple"),

      border = "white")


# — 3. GRAPH 2: Alcohol vs Quality (Boxplot) ——————————————

boxplot(alcohol ~ quality, data = wine,

      main   = "Alcohol Content by Quality",

      xlab   = "Quality Score",

      ylab   = "Alcohol (%)",

      col    = "lightblue",

      border = "steelblue")


# — 4. GRAPH 3: Correlation Heatmap ——————————————————————

cor_matrix <- cor(wine)


# Color palette

col_palette <- colorRampPalette(c("blue", "white", "red"))(20)


image(1:ncol(cor_matrix), 1:nrow(cor_matrix),
```

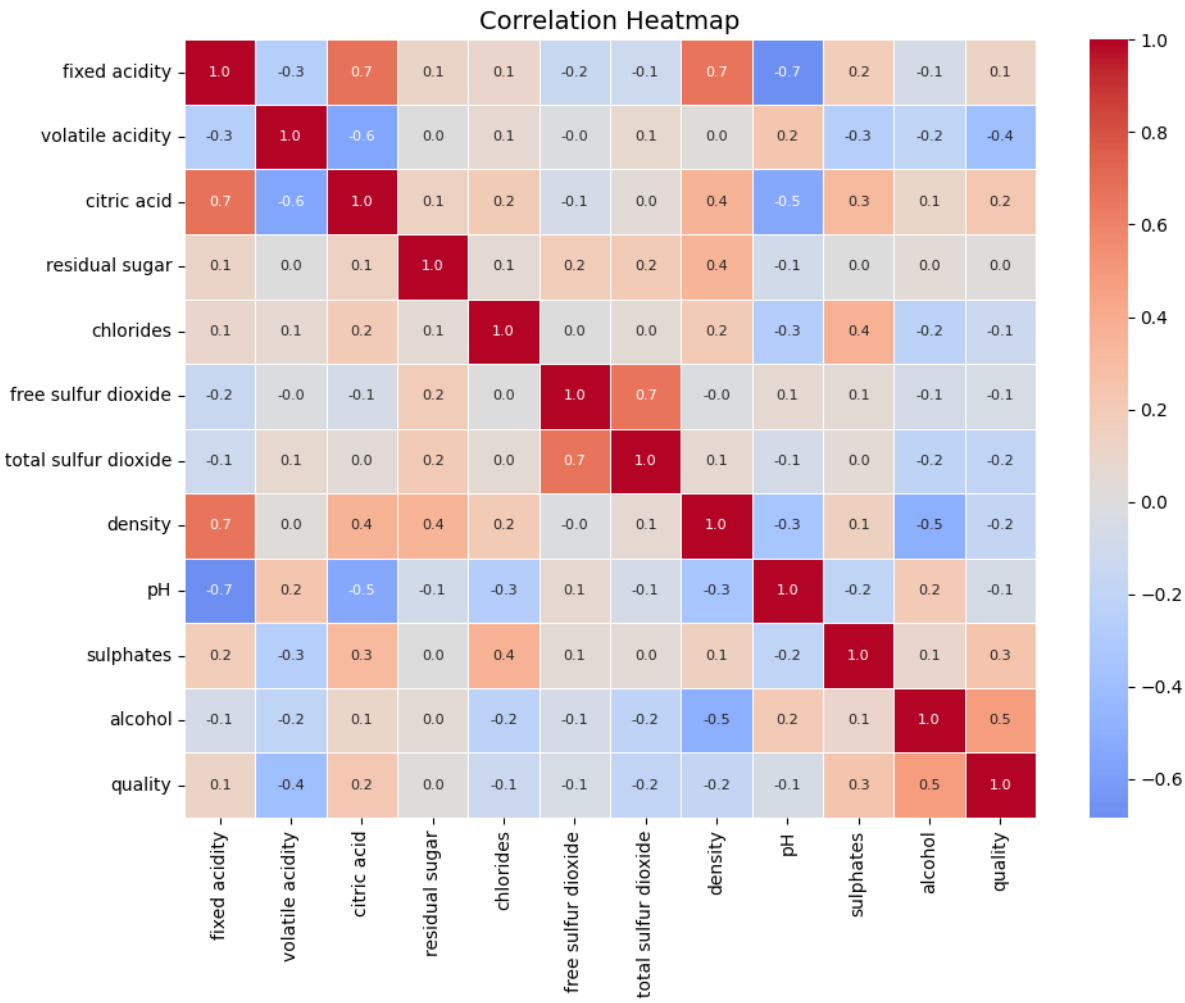
```
    t(cor_matrix[nrow(cor_matrix):1, ]),
    col  = col_palette,
    axes = FALSE,
    main = "Correlation Heatmap")

axis(1, at = 1:ncol(cor_matrix),
   labels = colnames(cor_matrix), las = 2, cex.axis = 0.6)
axis(2, at = 1:nrow(cor_matrix),
   labels = rev(rownames(cor_matrix)), las = 1, cex.axis = 0.6)

# Add correlation values
for (i in 1:nrow(cor_matrix)) {
  for (j in 1:ncol(cor_matrix)) {
    text(j, nrow(cor_matrix) - i + 1,
       round(cor_matrix[i, j], 1), cex = 0.5)
  }
}

# — 5. GRAPH 4: Volatile Acidity vs Quality (Boxplot) ————
boxplot(volatile.acidity ~ quality, data = wine,
       main   = "Volatile Acidity by Quality",
       xlab   = "Quality Score",
       ylab   = "Volatile Acidity",
       col    = "salmon",
       border = "darkred")

# — 6. GRAPH 5: Histogram of Alcohol Content ——————————
hist(wine$alcohol,
```

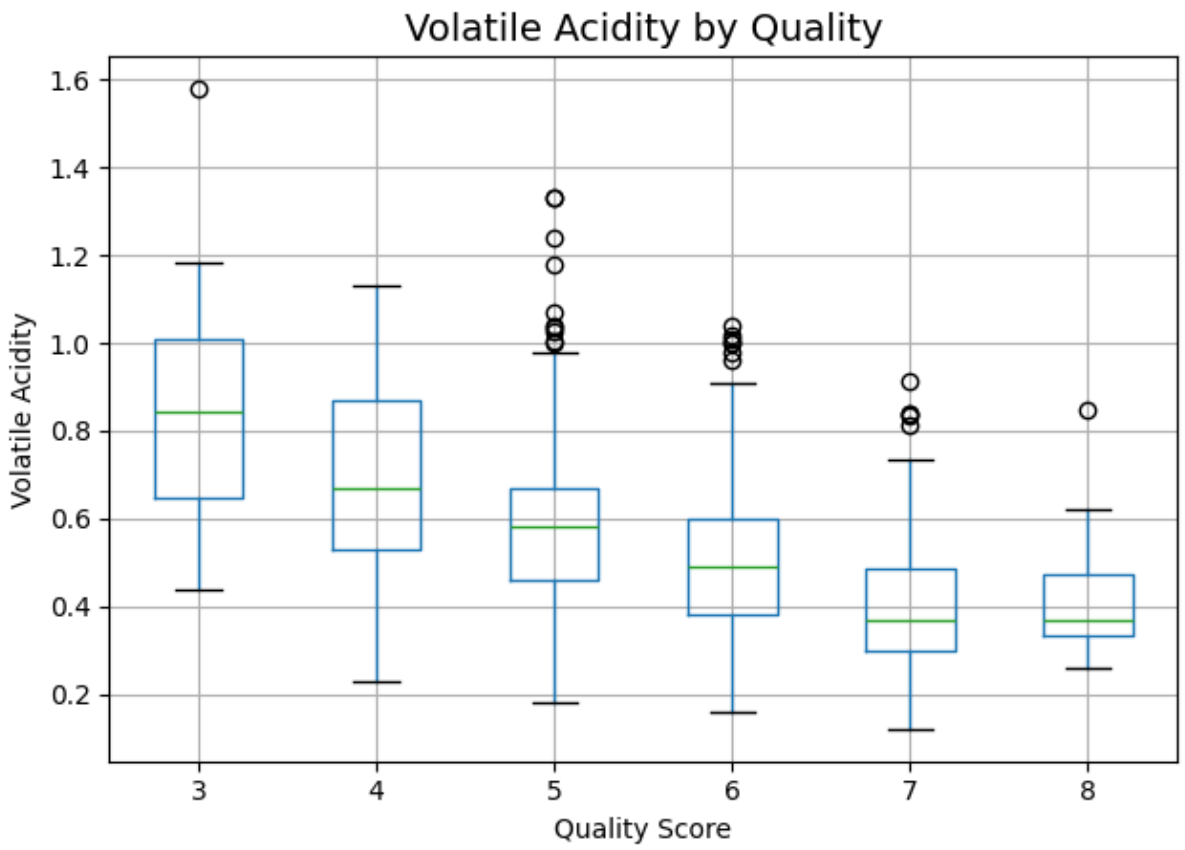
```
    main   = "Distribution of Alcohol Content",
    xlab   = "Alcohol (%)",
    ylab   = "Frequency",
    col    = "lightgreen",
    border = "darkgreen",
    breaks = 20)
abline(v = mean(wine$alcohol), col = "red", lwd = 2, lty = 2)
legend("topright", legend = paste("Mean =", round(mean(wine$alcohol), 2)),
      col = "red", lwd = 2, lty = 2)

# — 7. GRAPH 6: Scatter Plot – Alcohol vs Volatile Acidity ——
plot(wine$alcohol, wine$volatile.acidity,
    main  = "Alcohol vs Volatile Acidity",
    xlab  = "Alcohol (%)",
    ylab  = "Volatile Acidity",
    col   = ifelse(wine$quality >= 7, "green", "red"),
    pch   = 16,
    alpha = 0.5)
legend("topright",
      legend = c("Good Quality (>=7)", "Not Good (<7)"),
      col    = c("green", "red"),
      pch    = 16)

# — 8. BASIC MODEL ————————————————————————————————————
model <- lm(quality ~ ., data = wine)
summary(model)

predictions <- predict(model, wine)
```

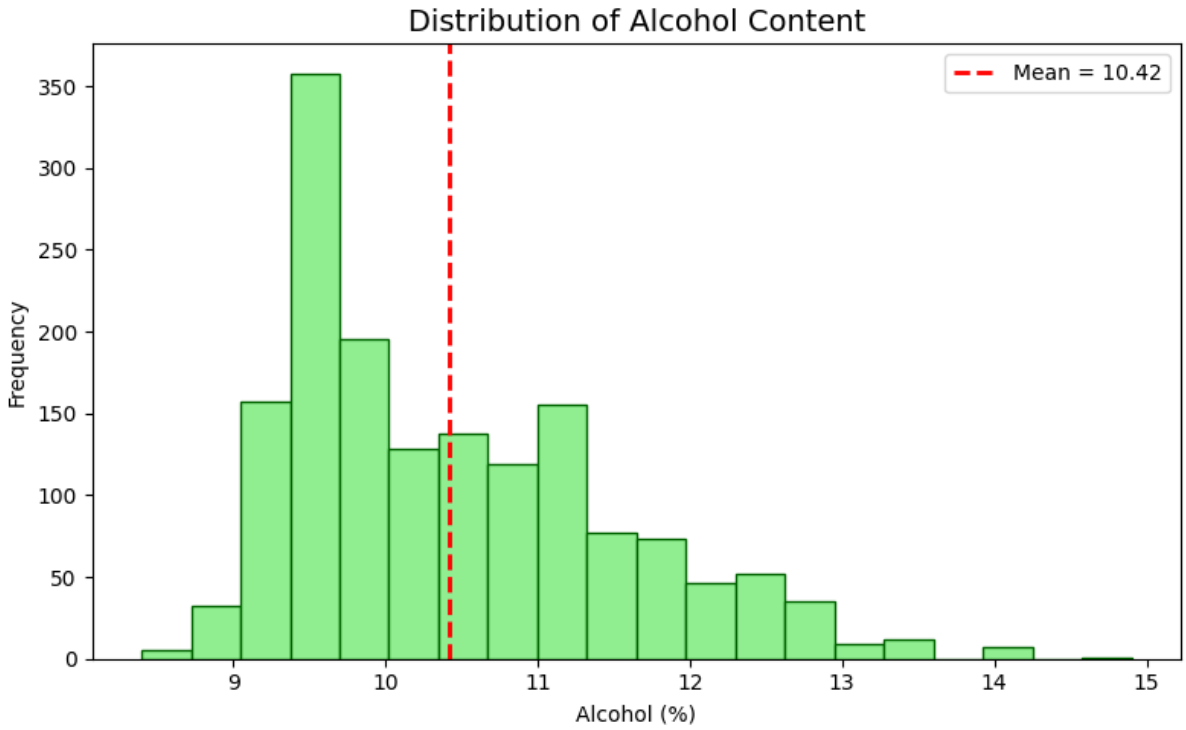
```
rmse <- sqrt(mean((predictions - wine$quality)^2))

cat("RMSE:", round(rmse, 4), "\n")
```

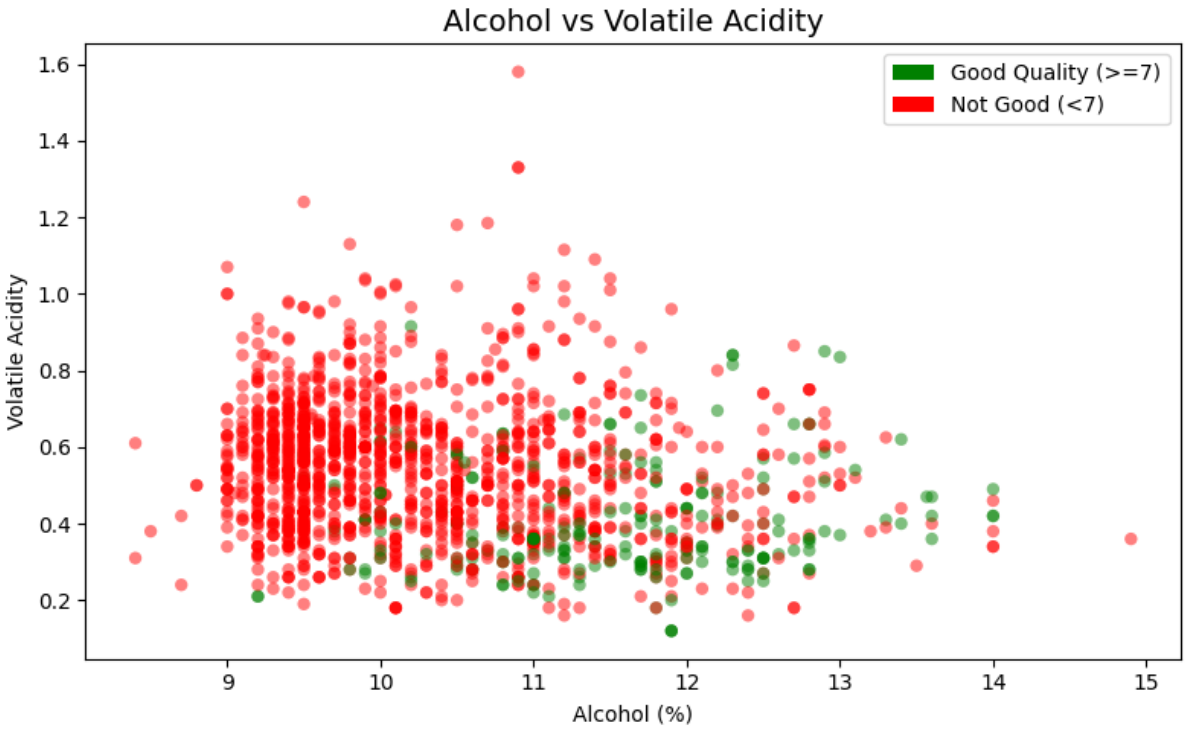





Correlation Heatmap
fixed acidity
volatile acidity
citric acid
residual sugar
chlorides
free sulfur dioxide
total sulfur dioxide
density
pH
sulphates
alcohol
quality
1.0 -0.3 0.7 0.1 0.1 -0.2 -0.1 0.7 -0.7 0.2 -0.1 0.1
-0.3 1.0 -0.6 0.0 0.1 -0.0 0.1 0.0 0.2 -0.3 -0.2 -0.4
0.7 -0.6 1.0 0.1 0.2 -0.1 0.0 0.4 -0.5 0.3 0.1 0.2
0.1 0.0 0.1 1.0 0.1 0.2 0.2 0.4 -0.1 0.0 0.0 0.0
0.1 0.1 0.2 0.1 1.0 0.0 0.0 0.2 -0.3 0.4 -0.2 -0.1
-0.2 -0.0 -0.1 0.2 0.0 1.0 0.7 -0.0 0.1 0.1 -0.1 -0.1
-0.1 0.1 0.0 0.2 0.0 0.7 1.0 0.1 -0.1 0.0 -0.2 -0.2
0.7 0.0 0.4 0.4 0.2 -0.0 0.1 1.0 -0.3 0.1 -0.5 -0.2
-0.7 0.2 -0.5 -0.1 -0.3 0.1 -0.1 -0.3 1.0 -0.2 0.2 -0.1
0.2 -0.3 0.3 0.0 0.4 0.1 0.0 0.1 -0.2 1.0 0.1 0.3
-0.1 -0.2 0.1 0.0 -0.2 -0.1 -0.2 -0.5 0.2 0.1 1.0 0.5
0.1 -0.4 0.2 0.0 -0.1 -0.1 -0.2 -0.2 -0.1 0.3 0.5 1.0
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6


Volatile Acidity by Quality
Volatile Acidity
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
3
4
5
6
7
8
Quality Score


Distribution of Alcohol Content
Mean = 10.42
Frequency
350
300
250
200
150
100
50
0
9
10
11
12
13
14
15
Alcohol (%)


Alcohol vs Volatile Acidity
Good Quality (>=7)
Not Good (<7)
Volatile Acidity
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
9
10
11
12
13
14
15
Alcohol (%)